SOFTWARE ENGINEER

Brandon Temple

Software engineer and M.S. candidate building automation-focused desktop, web, and data systems with C#, Python, Perl, SQL, JavaScript, and API integrations. Uses an AI-first engineering workflow with human architectural judgment, systems fundamentals, data structures, and code review to maintain correctness, performance, and long-term maintainability.

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TECHNICAL SKILLS

Languages C#, C++, C, Java, Python, TypeScript, JavaScript, HTML, SQL, Swift, Kotlin
Frameworks .NET 8/9, ASP.NET Core, WPF, WinUI 3
Databases SQL Server, PostgreSQL, EF Core
Libraries WebView2, EPPlus, ExcelJS
Tools Git, Visual Studio, VS Code, Postman, MSTest, CSS, SASS
Methods OOP, Data Structures, Algorithms, REST APIs, Event-Driven Architecture, TDD, Agile, AI-First Engineering, Architectural Judgment, Machine Learning, Address Resolution

PROFESSIONAL EXPERIENCE

Software Engineer (Data Systems) / Technology Coordinator

MSU Libraries, Mississippi State University | Starkville, MS | Aug 2024 - Present

- Built and maintained C#, Python, Perl, SQL, JavaScript, and HTML tools for MSU Libraries to automate reporting, file processing, metadata extraction, data export, and operational workflows.
- Developed Query, a browser-based reporting tool with configurable filters, API-backed query payloads, report history, and Excel export for live library item data.
- Built MetaTable, a WPF metadata scanner that processes directory trees, extracts file attributes, tracks progress in real time, and exports structured results to Excel.
- Automated repetitive reporting and data-preparation workflows that previously required manual collection, cleanup, formatting, and staff handoff.
- Integrated SirsiDynix APIs, SQL-backed data sources, and Excel export libraries to support internal reporting and library systems operations.
- Applied an AI-first engineering workflow with C#, C++, data structures, and systems fundamentals to review generated code for correctness, algorithmic efficiency, memory behavior, and maintainability.

PROJECTS

MetaTable

WPF File Metadata Scanner | C#, .NET 9, EPPlus

Created as part of my MSU Libraries role to scan directories, capture detailed file metadata, track progress in real time, and export results to Excel.
github.com/bt1142msstate/MetaTable

HTMLConverter

WinUI 3 Batch PDF Converter | .NET 8, WebView2

Developed a Windows app that converts HTML files, folders, and ZIP archives to PDF with drag-and-drop intake, batch processing, scaling controls, and automatic output archiving.
github.com/bt1142msstate/HTMLConverter

Query

Browser-Based Report Builder | JavaScript, HTML, CSS, ExcelJS

Created for MSU Libraries to create library item reports with configurable filters, structured query payloads, history tracking, and Excel export for large datasets.
github.com/bt1142msstate/Query

Ady Resolver

Mississippi Address Resolution Toolkit | Python, Machine Learning

Built a Python address-resolution toolkit and local browser app for resolving messy Mississippi addresses using cached public source data, typo-heavy evaluation data, and a staged matching pipeline.

github.com/bt1142msstate/Ady-Resolver

connect4-ai

Browser-Based Game AI Demo | JavaScript, Minimax, Alpha-Beta Pruning

Built a playable Connect 4 app with minimax, alpha-beta pruning, replay tooling, experiment modes, and headless validation for browser-based game AI.

github.com/bt1142msstate/connect4-ai

EDUCATION

Master of Science in Software Engineering

Mississippi State University | May 2027

Starkville, MS | GPA: 4.0/4.0

Bachelor of Science in Computer Science, Cum Laude

Mississippi State University | May 2024

Starkville, MS | GPA: 3.68/4.0

Honors: Phi Theta Kappa, Gamma Beta Phi

Associate of Arts in Computer Science

East Central Community College | May 2021

Decatur, MS | GPA: 3.85/4.0

Honors: Phi Theta Kappa, Gamma Beta Phi

President's List: Fall 2019, Spring 2020

Dean's List: Fall 2020

C++ Programming Award

Outstanding ECCC CS Student Award, awarded by Mississippi State University (custom cowbell)

ADDITIONAL EXPERIENCE

Community Assistant

College View, Mississippi State University | 2022 - 2024

- Supported student residents through communication, policy enforcement, community operations, and incident response.
- Coordinated resident support workflows while balancing graduate-level software engineering coursework.

STEM Leader & Math Tutor

Afternoon Adventure Learning Center | 2018 - 2021

- Tutored students in math, science, and technology.
- Prepared students for exams and improved study strategies.
- Designed engaging STEM activities with staff.

Computer Science Lab Monitor

East Central Community College | 2019 - 2020

- Maintained and troubleshot lab computers and systems.
- Assisted students with technical issues and software use.
- Supported problem solving and computer literacy.